

DSA Time Complexity MCQ

Questions & Answers

Q1. Which notation is commonly used to represent the best-case time complexity of an algorithm?

Answer: Ω (Omega) notation.

Q2. What does $O(1)$ represent in Big-O notation?

Answer: Constant-time complexity.

Q3. Which of the following asymptotic notations is the worst of all? ($O(n!)$, $O(n^3)$, $O(n^2)$, $O(1)$)

Answer: $O(n!)$ — factorial-time grows faster than cubic, quadratic, or constant.

Q4. How is a quadratic-time function expressed in Big-O notation?

Answer: $O(n^2)$.

Q5. Time complexity: `while(count>=1){count=count/3;}`

Answer: $O(\log n)$.

Q6. What does Big-O asymptotic notation provide?

Answer: An upper bound idea (worst-case growth).

Q7. Machine-independent way to compare algorithms?

Answer: Count the number of elementary/key operations as a function of input size.

Q8. Which method is commonly used to solve $T(n) = T(n/2) + O(1)$?

Answer: All of the mentioned options (Substitution, Recursion Tree, Master Theorem).

Q9. $S(1) = 6$, $S(n+1) = 4n + S(n)$. Order of growth?

Answer: $O(n^2)$.

Q10. `for(x=N; x>0; x--) for(y=1; y<=x; y++)`

Answer: $O(n^2)$.

Q11. In Big-O notation, what is $O(N)$ used to represent?

Answer: Linear-time complexity.

Q12. $T(n) = 4T(n/2) + O(n)$

Answer: $O(n^2)$ (Master Theorem).

Q13. `for(p=m; p>=m/3; p--)` with inner loop `q *= 3`

Answer: $O(n \log n)$.

Q14. Nested loops: `for(a=n; a>=1; a--) for(b=a; b>=1; b--) for(c=1; c<=50; c++)`

Answer: $O(n^2)$, because the innermost loop is constant (50 iterations).

Q15. Which notation is commonly used for average-case complexity?

Answer: Θ (Theta) notation.